

ADARSH R SHENOY

adarshrshenoy1@gmail.com | [Portfolio](#) | [Github](#) | [LinkedIn](#)

ABOUT ME

Robotics & AI Engineer specializing in **practical Deep Learning** application and system development. Proven ability to translate complex models into real-world tools across diverse domains (healthcare, aviation and retail). Key projects include **cerebAI**, a high-performance **ConvNeXt-based** stroke detection system designed for rapid (<10 second) clinical prototyping, deployed using **Streamlit** and **PyTorch**.

SKILLS AND ABILITIES

Core AI & Deep Learning:

PyTorch | TensorFlow | Scikit-learn | Deep Learning | Neural Networks | NLP | Computer Vision (OpenCV) | ConvNeXt | Explainable AI (Integrated Gradients)

Data Science & Analysis:

NumPy | Pandas | Matplotlib | Data Preprocessing | Model Evaluation

Programming:

Python | Java | JavaScript | SQL

Deployment & Tools:

Streamlit | Hugging Face | Git | Linux | Jupyter Notebook | Postman | Docker (basic)

Web Development (supporting):

HTML | CSS | PHP (basic)

Languages:

English [C1] | German [B1 – Beginner] | Hindi | Mandarin [A1 – Basic]

EXPERIENCE

Gladwell Academy India Pvt. Ltd. – Bangalore, India

12.24-04.25

AI/ML Intern

- Participated in a research-focused internship program exploring optimization-based AI use-cases under mentorship.
- Independently developed a Bangalore house price prediction ML model and showcased it as a proof-of-concept solution in the final project review.
- Contributed to knowledge-sharing sessions on improving model evaluation and deployment readiness.

EDUCATION

NMAM Institute of Technology

2022-2026

- B.Tech in Robotics & Artificial Intelligence
- CGPA: **8.68/10**
- Specialization: Artificial Intelligence

PROJECTS

cerebAI - [Link](#)

cerebAI is a **deployed AI-based stroke detection** system built using the **ConvNeXt Base** architecture in PyTorch to classify CT scans into *No Stroke*, *Ischemic*, or *Hemorrhagic*. It achieved state-of-the-art diagnostic performance with a mean IoU of **0.9843** and a weighted F1 Score of **0.9805**, confirming high reliability. To ensure explainability, **Integrated Gradients** (Captum) was implemented for pixel-level attributions, highlighting pathology regions and providing transparent clinical insights. The pipeline supports professional **DICOM** preprocessing, including Hounsfield Unit conversion and windowing, as well as compatibility with JPG, JPEG, and PNG formats. **Deployed on Hugging Face Spaces** via **Streamlit SDK**, the system delivers rapid, real-time results making it suitable for **clinical prototyping** and medical decision support.

LunarisAI - [Link](#)

A **menstrual cycle prediction system** using machine learning to estimate the **next period date** based on **user data**. Built with Random Forest Regressor and deployed via **Streamlit**. Adapts to inputs like stress, PCOS, and birth control to provide **personalized predictions**.

House price prediction system - [Link](#)

Developed a **machine learning model** to predict house prices in Bangalore. Achieved **83–86% accuracy**. Built to **assist small-scale real estate businesses** with **property valuation**.

Flight Customer Satisfaction System - [Link](#)

Built a **logistic regression model** to predict customer satisfaction based on **post-flight feedback**. Analysed factors such as service quality, seat comfort, and overall experience to classify users as **satisfied or dissatisfied**. Achieved **76–80% accuracy**.

Direct kinematics for robots - [Link](#)

A **mathematical tool** for computing the direct **kinematics of robotic arms**. Helps users determine the robot's **position** and **orientation** within a 3D space.

AI-NPC Interaction Engine - [Link](#)

A modular Python **engine simulating** intelligent **NPC interactions** using OOP and **ML**. Includes intent classification, **emotion-aware responses**, and turn-based mechanics.

Bookstore website - [Link](#)

A **full-stack development project** (in progress). Aims to create a **functional** online **bookstore** with modern user-facing and backend features.